Project Title	**Smart Logistics Management & Analytics Platform**
Skills take away From This Project	**Python scripting, Data Collection, Data Management using SQL, Streamlit**
Domain	**Logistics**

Problem Statement:

In modern logistics operations, companies similar to **DHL** and **FedEx** manage thousands of shipments daily across multiple routes, warehouses, and courier personnel.

Operational inefficiencies such as delivery delays, high transportation costs, underutilized warehouses, and inconsistent courier performance can significantly impact profitability and customer satisfaction.

However, raw operational data is often scattered across multiple files and systems (shipment records, tracking logs, cost sheets, route details). Without centralized analytics, it becomes difficult for management to:

- Monitor delivery performance
- Identify bottleneck routes
- Track real-time shipment status

- Analyze operational costs
- Evaluate courier efficiency
- Detect cancellation patterns

This project aims to build a centralized **Smart Logistics Management and Analytics Platform** that consolidates operational data into a MySQL database and provides an interactive Streamlit dashboard for real-time analysis and decision-making.

2️⃣ Project Objective

The objective of this project is to design and implement an end-to-end logistics analytics system that:

1. Ingests large-scale logistics datasets (70,000+ shipment records).
2. Stores data in a normalized MySQL relational database.
3. Enables shipment-level tracking through status logs.
4. Provides operational insights via Streamlit dashboards.
5. Supports filtering, KPI monitoring, and business performance evaluation.

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3️⃣ Data Architecture Overview

The system integrates structured data across six core datasets:

- Shipments
- Shipment Tracking
- Courier Staff

- Routes
- Warehouses
- Costs

These datasets simulate real-world logistics operations.

4️⃣ Database Schema (MySQL)

DATASET LINK - 📂 Logistics_dataset

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Complete Table Schema with Column Descriptions

📦 Table: `shipments`

Column Name	Data Type	Description
shipment_id	VARCHAR(50) PRIMARY KEY	Unique identifier for each shipment/order
order_date	DATE	Date when the shipment order was created
origin	VARCHAR(100)	City/location where shipment starts

destination	VARCHAR(100)	Delivery city/location
weight	DECIMAL(10,2)	Weight of the shipment in kg
courier_id	VARCHAR(50)	Courier responsible for delivery (FK to courier_staff)
status	VARCHAR(50)	Current shipment status (Delivered, In Transit, Cancelled, etc.)
delivery_date	DATE NULL	Date when shipment was delivered (NULL if not delivered yet)

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Table: `shipment_tracking`

Column Name	Data Type	Description
tracking_id	INT PRIMARY KEY	Unique identifier for tracking event
shipment_id	VARCHAR(50)	Shipment linked to this event (FK to shipments)

status	VARCHAR(50)	Status update at this stage (Picked Up, In Transit, Delivered, etc.)
timestamp	DATETIME	Date & time when this tracking event occurred

Table: `courier_staff`

Column Name	Data Type	Description
courier_id	VARCHAR(50) PRIMARY KEY	Unique identifier for courier employee
name	VARCHAR(150)	Full name of courier
rating	DECIMAL(3,1)	Courier performance rating (1–5 scale)
vehicle_type	VARCHAR(50)	Type of delivery vehicle used (Bike, Van, Truck, Car)

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Table: `routes`

Column Name	Data Type	Description

route_id	VARCHAR(50) PRIMARY KEY	Unique identifier for transport route
origin	VARCHAR(100)	Starting city/location of route
destination	VARCHAR(100)	Ending city/location of route
distance_km	DECIMAL(10,2)	Distance between origin and destination in kilometers
avg_time_hours	DECIMAL(5,2)	Average travel time expected for this route

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Table: `warehouses`

Column Name	Data Type	Description
warehouse_id	VARCHAR(50) PRIMARY KEY	Unique identifier for warehouse
city	VARCHAR(100)	City where warehouse is located
state	VARCHAR(50)	State or region of warehouse

capacity	INT	Maximum shipment capacity warehouse can handle

Table: `costs`

Column Name	Data Type	Description
shipment_id	VARCHAR(50) PRIMARY KEY	Shipment linked to cost record (FK to shipments)
fuel_cost	DECIMAL(15,2)	Fuel cost incurred for shipment delivery
labor_cost	DECIMAL(15,2)	Courier labor cost for shipment
misc_cost	DECIMAL(15,2)	Additional operational costs (handling, packaging, etc.)

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5️⃣ System Workflow

1. Import CSV/JSON datasets into MySQL.
2. Establish foreign key relationships.
3. Build optimized SQL queries for KPIs.

4. Connect Streamlit to MySQL using a Python database connector.
5. Display real-time data with filtering options.

6️⃣ Streamlit Dashboard Features

The Streamlit interface will include:

🔎 A. Shipment Search & Filtering

- Search by shipment_id
- Filter by:
  - Status (Delivered, Cancelled, In Transit)
  - Origin / Destination
  - Date range
  - Courier

📊 B. Operational KPIs

- Total Shipments
- Delivered Shipments %
- Cancelled Shipments %
- Average Delivery Time
- Total Operational Cost

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📈 C. Analytical Views

1️⃣ Delivery Performance Insights

- Average delivery time per route
- Most delayed routes
- Delivery time vs distance comparison

2️⃣ Courier Performance

- Shipments handled per courier
- On-time delivery %
- Average rating comparison

3️⃣ Cost Analytics

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- Total cost per shipment
- Cost per route
- Fuel vs labor percentage contribution
- High-cost shipments

4️⃣ Cancellation Analysis

- Cancellation rate by origin
- Cancellation rate by courier
- Time-to-cancellation analysis

5️⃣ Warehouse Insights

- Warehouse capacity comparison
- High-traffic warehouse cities

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7. Potential Business Insights

This system can help management answer:

- Which routes have the highest delays?
- Which couriers handle most shipments?
- Are high-rated couriers delivering faster?
- Which shipments are most expensive?
- Is cost directly proportional to weight?
- Which cities generate maximum cancellations?
- Which routes are underperforming relative to distance?

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8. Expected Outcome

A fully functional MySQL-backed Streamlit application that provides:

- Real-time shipment tracking
- Operational KPI monitoring
- Performance benchmarking
- Cost transparency
- Actionable logistics insights

The system mimics enterprise-grade logistics analytics platforms used by global delivery organizations.

Results:

The expected outcome is a fully functional web application built with Streamlit capable of analyzing logistics data.

Skills Takeaway:

1. **Data Extraction:** Json Parsing, working with CSV
2. **SQL Database Management:** Designing tables and schema.
3. **Data Analysis:** Writing SQL queries to derive insights.

Project Evaluation metrics:

1. **Data Extraction Accuracy:**Successful parsing of the desired data from json and csv sources without errors or missing values.
2. **SQL Database Design:**Proper structuring of tables with normalized data and correct relationships between them.
3. **Query Efficiency:**Ability to write optimized SQL queries that retrieve relevant insights quickly.
4. **Streamlit Application Functionality:**Smooth navigation and interactive features for data display and analysis.
5. **Project Completeness:** End-to-end workflow execution from data extraction to SQL storage and application using streamlit.
6. **Documentation Quality:**Clearly upload all files in github.
7. **Presentation and Usability:**An intuitive user interface with meaningful outputs and actionable insights.
8. **Error Handling:**Ability to manage database constraints, or user input errors effectively.
9. **Innovation and Creativity:**Unique implementation ideas, or insightful SQL queries.

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Technical Tags:

1. **Languages**: Python
2. **Database**: MySQL/PostgreSQL
3. **Application**: Streamlit

** ***In github, upload all your code files. Submit your repository link in the ZEN portal in the Project Section. Also , you have to present your project with one of our mentors in a live session. Refer to the form provided at the end of the document.***

Project Guidelines:

1. **Coding Standards**
    - Use meaningful names: Variables, functions, and database tables should have descriptive names.
    - Follow PEP 8 (for Python): Maintain consistent formatting with proper indentation and spacing.
    - Modularize your code: Break your code into functions or classes to enhance readability and reusability.
    - Error handling: Implement try-except blocks for handling API errors and SQL exceptions.
    - Document your code: Include docstrings and comments to explain logic and functions.
2. **SQL Database Practices**
    - Normalize tables: Avoid redundancy and ensure efficient data storage.
    - Use indexes: Optimize query performance with appropriate indexing.
    - Follow naming conventions: Use consistent and descriptive names for tables and fields.
3. **Streamlit Application Development**
    - Interactive features: Ensure the UI is responsive, with interactive widgets for filters.
    - Minimalist design: Keep the layout simple for a smooth user experience.
    - Performance optimization: Avoid loading all data at once—use pagination or batch processing where possible.
4. **General Best Practices**
    - Test frequently: Regularly test each component (e.g., API requests, SQL queries, Streamlit app) during development.
    - Backup your data: Maintain backups of your SQL database and code.
    - Documentation: Provide a README file with setup instructions, project objectives, and a demo walkthrough.

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Reference:

*****If you don't know how to approach the project, kindly refer to the project orientation recording provided in this table. (Available in English)***

Streamlit Doc	https://docs.streamlit.io/library/api-reference
Streamlit recording (English)	Intel_AIML_Session on Streamlit [ENGL…
Streamlit reference Code	IntelStreamlit2.ipynb
Project Live Evaluation Metrics	Project Live Evaluation
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
Project Orientation (English)	Project Orientation Session : First_Proje…

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Timeline:

The project must be completed and submitted within **10 days from the assigned date.**

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.
Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday to Saturday (3:30PM to 4:30PM)

Session link : Project Doubt Session [DS/AIML]

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open on Saturday and Sunday Only on Every Week

Timing: Monday-Saturday (5:30 PM to 6:30 PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA

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